

B.Tech. Project
On
ECG Analysis Using ML

Report submitted in partial fulfillment of the requirements for the
B. Tech. degree in Instrumentation and Control Engineering
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CANDIDATES DECLARATION



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We, Mayank Kumar (2022UIC3598), Ritik Jain (2022UIC3603), Akshay Sharma (2022UIC3605) and Shanu Yadav (2022UIC3635) students of B. Tech., Department of Instrumentation and Control Engineering, hereby declare that the Project-Thesis titled “ECG Analysis using ML ” which is submitted by us to the Department of Instrumentation and Control Engineering, Netaji Subhas University of Technology (NSUT) Dwarka, New Delhi in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology is our original work and not copied from any source without proper citation. The manuscript has been subjected to plagiarism check by Turnitin software. This work has not previously formed the basis for the award of any other Degree

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CERTIFICATE



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This is to certify that the work embodied in project thesis titled, “ECG Analysis using ML” by Mayank Kumar (2022UIC3598), Ritik Jain (2022UIC3603), Akshay Sharma (2022UIC3605) and Shanu Yadav (2022UIC3635) is the Bonafide work of the group submitted to Netaji Subhas University of Technology for consideration in 8th Semester B.Tech. Project Evaluation.

The original Research work was carried out by the team under my/our guidance and supervision in the academic year 2025-2026. This work has not been submitted for any other diploma or degree of any university. Plagiarism has been checked and is under the prescribed limit. On the basis of the declaration made by the group, we recommend the project report for evaluation.

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ABSTRACT

Cardiovascular diseases (CVD) remain a leading cause of global mortality, and early detection from ECG signals is essential for timely intervention. However, accurate interpretation of multi-lead ECGs is complex and traditionally requires expert cardiologists. This project aims to develop an end-to-end deep learning pipeline for automatic diagnosis of cardiac abnormalities using 12-lead ECG recordings, following the methodology provided in the open-source framework “Automatic ECG Diagnosis”.

The dataset consists of standardized ECG records preprocessed into $4096 \text{ samples} \times 12 \text{ leads}$, enabling consistent input for model training. The workflow begins with exploratory data analysis (EDA) to understand signal characteristics, identify noise patterns, and verify label distributions across the ten target cardiac conditions: 1st-degree AV Block, Right Bundle Branch Block, Left Bundle Branch Block, Sinus Bradycardia, Atrial Fibrillation, and Sinu Tachycardia, ST-Elevation, Premature Ventricular Contraction, Ventricular Tachycardia, Wolff-Parkinson-White.

Preprocessing includes signal normalization, padding, and lead wise organization to match the model's architecture. A 1D Residual Convolutional Neural Network is implemented to learn temporal dependencies in ECG waveforms. The model is evaluated using clinically relevant metrics such as F1-score, Precision, and Recall.

Performance is assessed using separate training, validation, and test splits following the recommended evaluation protocol of the repository. Results demonstrate that the residual CNN effectively captures morphological ECG patterns and produces reliable multi-label predictions. The framework achieves competitive classification performance across the ten cardiac abnormalities, highlighting the potential of deep learning to support automated ECG screening. This project establishes a scalable foundation for future improvements involving real-time inference, advanced signal preprocessing, and clinical validation

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Chapter 1: Introduction

1.1 Problem Statement

Cardio-vascular diseases (CVDs) continue to be a major global health concern, with early detection playing a critical role in improving patient outcomes. Electrocardiography (ECG) is one of the most widely used, non-invasive diagnostic tools for detecting cardiac abnormalities, yet accurate interpretation typically requires clinical expertise and is prone to human error. As the volume of ECG recordings grows in modern healthcare systems, there is an increasing need for automated, scalable, and reliable diagnostic systems. From a technical perspective, developing deep learning models for ECG classification introduces several challenges. ECG signals are high-dimensional, consisting of 12 leads that capture complex temporal patterns across thousands of samples. The recordings often contain noise artifacts, including baseline drift, muscle movement interference, and electrical disturbances, which necessitate careful preprocessing. Furthermore, ECG datasets typically exhibit class imbalance, as abnormal rhythms such as Atrial Fibrillation or 1st-degree AV Block occur far less frequently than normal patterns. An additional challenge is the multi-label nature of ECG diagnosis, since a single recording may present more than one cardiac abnormality simultaneously. These challenges emphasize the need for a robust deep learning pipeline capable of extracting morphological features, handling noisy and imbalanced data, and producing accurate multi-label predictions.

1.2 Objectives

The primary objective of this project is to design and implement an end-to-end Deep Learning system for automated ECG-based diagnosis. The study focuses on the following specific aims:

- **1. ECG Data Standardization:** To build a preprocessing pipeline that filters noise, normalizes amplitude values, aligns sampling rates, and prepares the signals in a fixed 4096×12 format suitable for deep learning models.
- **2. Signal Understanding and Visualization:** To perform exploratory data analysis (EDA) to examine ECG waveform characteristics, class distributions, and notable morphological differences between normal and abnormal recordings.
- **3. Deep Learning Model Development:** To design and train a 1D Convolutional Neural Network capable of capturing temporal and morphological ECG features for multi-label classification across ten cardiac abnormalities.
- **4. Performance Evaluation:** To assess model performance using clinically relevant metrics such as Precision, Recall, F1-score, and confusion matrices for each abnormality.

1.3 Scope of the Study

The study uses raw 12-lead ECG signals obtained from a publicly available dataset, PTB-XL. The dataset includes 10-second recordings sampled at 400 Hz, annotated for ten clinically important cardiac abnormalities.

- **Inclusions:** The analysis covers only raw ECG signals and focuses solely on deep learning-based classification of the ten selected abnormalities.
- **Methodology:** It uses 1D Convolutional kernels (filters) that slide over the signal. These filters act as automated "feature detectors" for slopes, peaks, and segments.
- **Target Variable:** After the convolutions find the patterns, these final layers act like a "voting committee" to decide which abnormality the patterns represent.
- **Exclusions:** The study does not involve clinical trials, demographic analysis, tabular medical records, or imaging-based diagnostic tools (e.g., MRI/CT). The proposed system acts as a supplementary screening tool rather than a replacement for professional medical interpretation.

Chapter 2: Literature Review

2.1 Overview of ECG-Based Cardiac Abnormality Detection

Electrocardiography (ECG) is a primary diagnostic tool for identifying cardiac abnormalities such as arrhythmias, conduction blocks, and ischemic changes. A standard 12-lead ECG captures the heart electrical activity from multiple spatial perspectives, enabling clinicians to detect subtle deviations in waveform morphology. Although clinically well-established, manual ECG interpretation is limited by inter-observer variability, increasing workload, and the growing volume of ECG data in modern healthcare systems. Traditional automated ECG analysis systems rely on handcrafted feature extraction—such as QRS detection, heart rate variability, and time-frequency transformations. While effective for simple patterns, these rule-based approaches struggle with complex waveform variations and non-linear relationships inherent in pathological ECGs. With the rise of large ECG repositories and computational advances, deep learning has emerged as a promising approach capable of learning morphological patterns directly from raw signals, thereby outperforming classical feature-engineered models.

2.2 Existing Deep Learning Approaches for ECG Classification

- Recent years have witnessed substantial progress in the application of Machine Learning and Deep Learning to ECG diagnostics. These techniques are capable of processing high-dimensional waveform data while capturing subtle temporal dependencies.
- Convolutional Neural Networks (CNNs): CNNs have become the dominant architecture for ECG analysis due to their ability to extract local waveform features such as P-waves, QRS complexes, and T waves. Studies using 1D CNNs have shown strong performance in classifying arrhythmias from raw ECG signals, often exceeding traditional methods in accuracy and robustness.
- Residual Neural Networks (ResNet-based models): Modern ECG research increasingly relies on deep residual architectures, which enable very deep networks by incorporating skip connections that prevent vanishing gradients. The GitHub implementation used in this project follows this paradigm, demonstrating high performance across multiple cardiac abnormalities.
- Recurrent Neural Networks (RNNs) and LSTMs: Earlier studies explored RNNs and LSTM architectures to capture long-term dependencies in ECG sequences. While effective in modeling temporal context, these models tend to be computationally expensive and less scalable compared to CNN-based alternatives.
- Hybrid CNN-RNN Models: Some research combines convolutional layers for feature extraction with recurrent layers for sequential modeling. Though such hybrids improve accuracy, they also increase complexity and training time.

2.3 Limitations of Current Methods Limitations of Current Methods

Despite advancements, several practical limitations persist in existing ECG classification research:

- **The Class Imbalance Problem:** ECG datasets are naturally imbalanced, with normal rhythms greatly outnumbering pathological cases. This imbalance often causes models to favor majority classes, resulting in poor sensitivity for rare but clinically critical abnormalities
- **Many studies achieve high overall accuracy by exploiting class imbalance—**predicting most samples as “normal.” However, this leads to elevated False Negative rates, which are unacceptable in clinical settings where missing a diagnosis can have severe consequences.
- **Signal Noise and Recording Variability:** ECG signals frequently contain artifacts such as baseline drift, motion noise, electrode misplacement, and electrical interference. Models trained without proper preprocessing may overfit noise rather than meaningful cardiac features.
- **Limited Generalization Across Datasets:** Deep learning models often exhibit reduced performance when applied to datasets collected from different devices, sampling rates, or patient populations. This limits their real-world deployment without domain adaptation.
- **Many studies prioritize statistical metrics like accuracy or F1-score, while clinical priorities emphasize Recall (Sensitivity) to avoid missed diagnoses.** Few works explore threshold tuning or other techniques to explicitly balance prediction confidence with patient safety

Chapter 3: Data Acquisition and Understanding

3.1 Dataset Description

The dataset utilized in this study is PTB XL, widely used for research on automated ECG interpretation. These datasets contain standardized 12-lead ECG recordings sampled at 400 Hz and annotated for multiple cardiac abnormalities, providing an excellent foundation for training deep learning models. The specific subset of data used for this project consists of cleaned and preprocessed ECG signals that follow the input structure required by the model implementation.

- Source: PTB-XL dataset
- Sample Size: The dataset contains 20000 annotated ECG recordings depending on the subset used.
- Data Type: Raw 12-lead ECG waveforms stored as arrays of shape (4096×12) after preprocessing.
- Target Variable: 10 cardiac abnormalities—1st-degree AV Block, Right Bundle Branch Block, Left Bundle Branch Block, Sinus Bradycardia, Atrial Fibrillation, and Sinu Tachycardia, ST-Elevation, Premature Ventricular Contraction, Ventricular Tachycardia, Wolff-Parkinson-White.

3.2 Feature Dictionary

Unlike tabular medical datasets, ECG classification relies on waveform data rather than categorical and numerical tabular features. Therefore, the “features” in this context correspond to the 12 spatial leads that measure electrical activity from different anatomical perspectives of the heart.

The key signal components used for analysis are categorized below:

<i>Category</i>	<i>Feature Name</i>	<i>Description</i>	<i>Units/Encoding</i>
<i>ECG Leads</i>	Lead I	Electrical activity between RA–LA	mV
	Lead II	Electrical activity between RA–LL	mV
	Lead III	Electrical activity between LA–LL	mV
	aVR	Augmented right limb lead	mV
	aVL	Augmented left limb lead	mV
	aVF	Augmented foot lead	mV
<i>Precordial Leads</i>	V1	Chest lead capturing septal activity	mV
	V2	Chest lead capturing anterior-septal activity	mV
	V3	Transition zone lead	mV
	V4	Anterior left-ventricular activity	mV
	V5	Lateral left-ventricular activity	mV
	V6	Lateral wall activity	mV

Table 1

These waveform leads collectively encode morphological ECG features such as P-waves, QRS complexes, T waves, ST segments, and arrhythmic patterns essential for diagnostic classification.

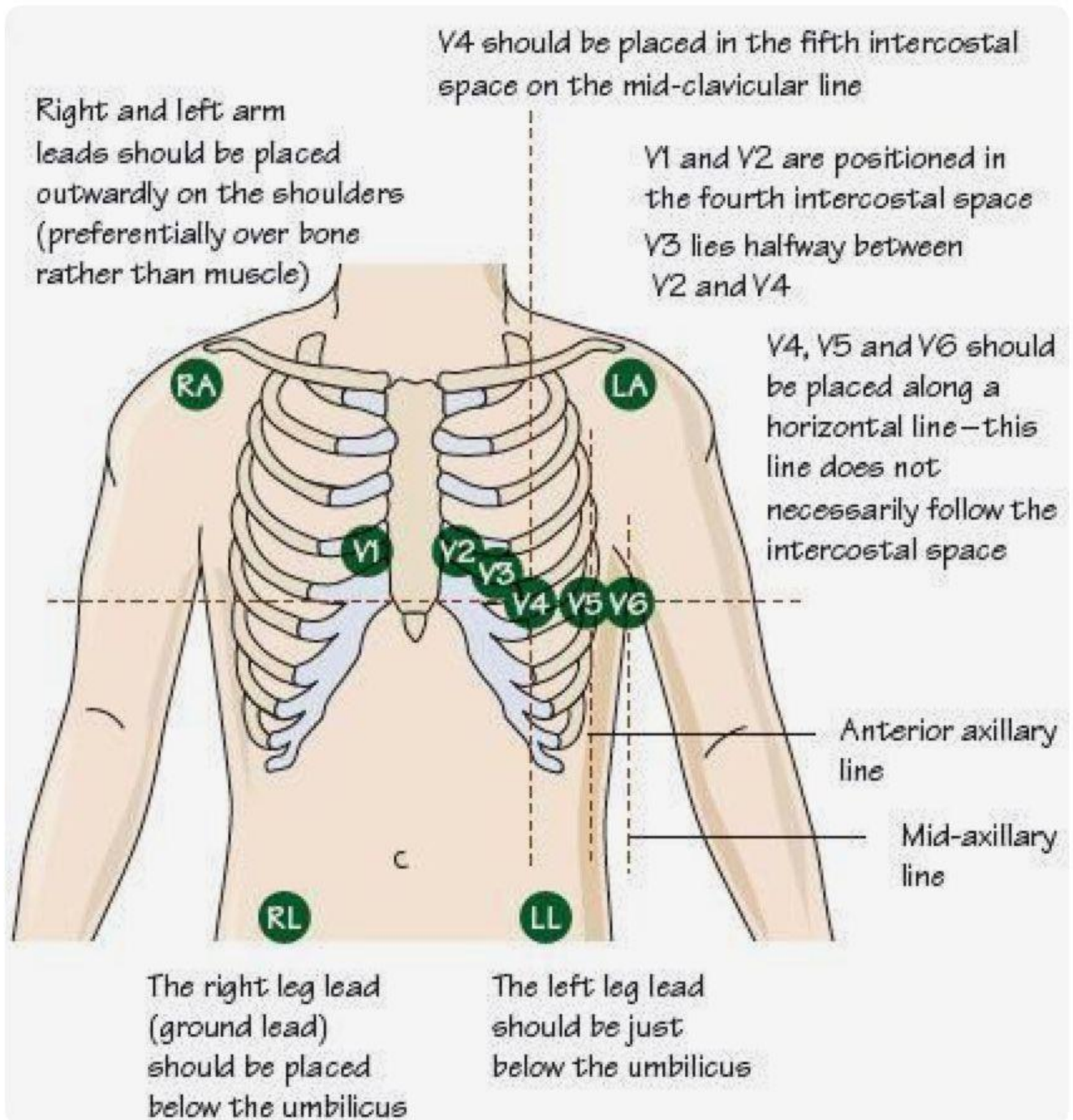


Fig 1

3.3 Initial Data Quality Assessment

A detailed Exploratory Data Analysis (EDA) was conducted to evaluate the suitability and consistency of the ECG signals prior to model training. Several important data quality considerations were identified:

1. **Signal Noise and Artifacts:** ECG recordings contain common distortions such as baseline wander, muscle noise, and powerline interference. Without proper filtering, these artifacts may obscure key waveform structures and mislead the deep learning model.
2. **Length Variability:** Some ECGs differ slightly in duration due to device variations or recording errors. To ensure uniform input size, all signals were resampled or padded to achieve a standardized shape of 4096 samples per lead.
3. **Class Imbalance:** The distribution of diagnostic labels showed notable imbalance among the ten target abnormalities. For example, conditions like Atrial Fibrillation or ST abnormalities occur less frequently than normal ECGs. This imbalance can bias the model toward majority classes unless corrected through weighted loss function
 - Normal Class: Typically constitutes the majority of samples.
 - Abnormal Class: Each abnormality represents a smaller proportion of total ECGs.
 - Implication: Due to the imbalance, standard accuracy would be an unreliable metric; a model predicting “Normal” for most recordings could falsely appear to perform well. This required the use of class-balanced training strategies and recall-focused evaluation to ensure clinical relevance.

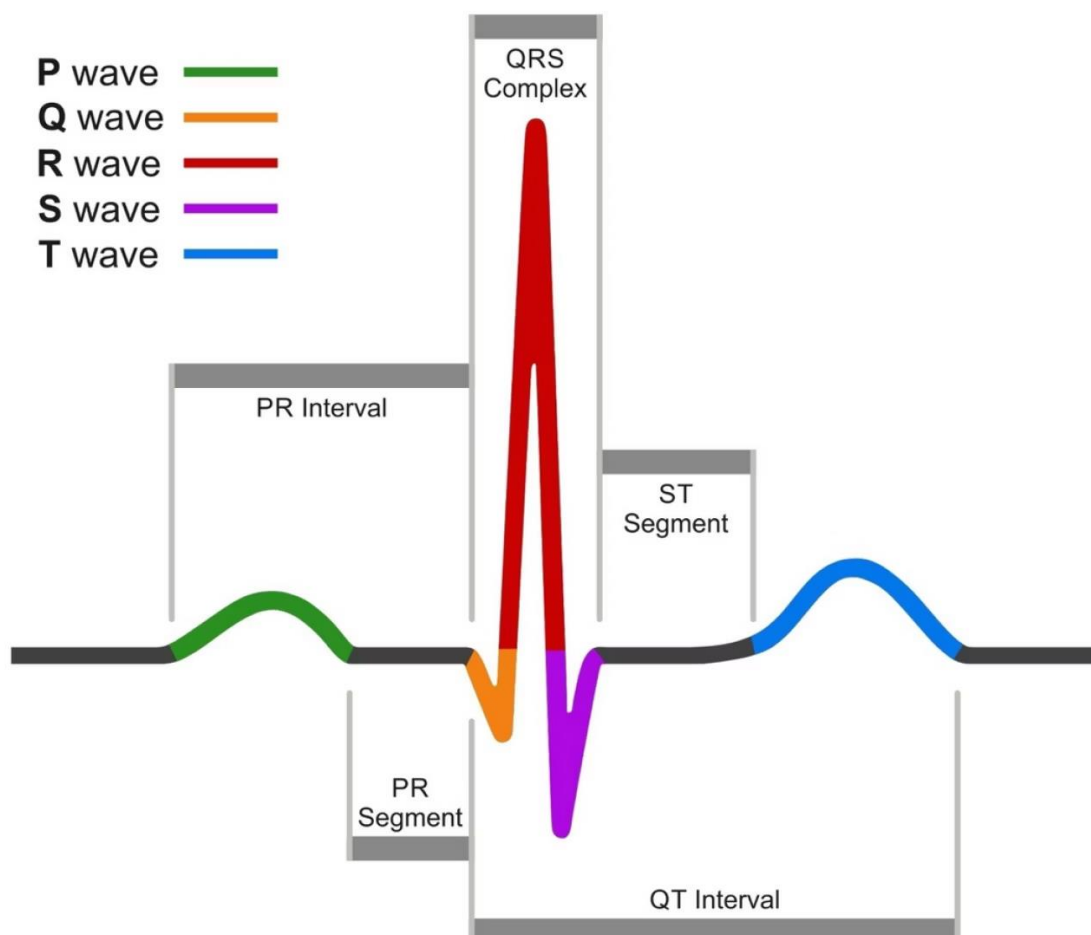


Fig 2

Chapter 4: Data Preprocessing and Feature Engineering

4.1 Data Cleaning & Handling Missing Values

Real-world ECG data is rarely perfect. Initial inspection revealed that certain recordings contained noise artifacts, inconsistent signal lengths, or missing waveform segments due to device-related interruptions during acquisition.

- **Noise Removal Strategy:** A combination of baseline wander removal and bandpass filtering was applied to eliminate low-frequency drift and high-frequency interference. This ensures that clinically relevant ECG components (P-QRS-T complexes) remain intact while removing distortions that may mislead the model.
- **Length Standardization:** Since a few ECG recordings varied slightly in duration, all signals were resampled or padded to achieve a consistent input shape of 4096 samples $\times$ 12 leads, preserving the original temporal characteristics required for deep learning.

4.2 Addressing Encoding and Structural Issues

During the preprocessing phase, several structural inconsistencies were identified across ECG records. These variations had the potential to introduce noise into the training process if left uncorrected:

- **Lead Ordering:** The raw dataset stored in ECG leads in different sequences depending on the recording device. A standardized clinical ordering (I, II, III, aVR, aVL, aVF, V1–V6) was enforced to ensure consistency across all samples.
- **Signal Polarity Issues:** This numerical inversion violates the ordinal assumption of many machine learning algorithms. A gradient-based model (like a Neural Network) would interpret "2" (No) as having a higher magnitude of the feature than "1" (Yes), logically inverting the risk signal.

4.3 Feature Selection & Redundancy Removal

In contrast to tabular datasets, ECG classification relies entirely on waveform morphology rather than manually engineered features. However, redundancy may still occur at the signal level. In contrast to tabular datasets, ECG classification relies entirely on waveform morphology rather than manually engineered features. However, redundancy may still occur at the signal level.

- **Redundancy Identified:**
 - Some recordings contained duplicated or corrupted leads due to acquisition errors.
 - A small subset exhibited flatline segments indicating electrode failure.
- **Action Taken:** Faulty or duplicated leads were removed or corrected through interpolation, while only clinically valid 12-lead waveforms were retained for training.

4.4 Data Scaling & Normalization

Signal normalization was applied to ensure that amplitude variations between recordings did not adversely affect gradient-based learning.:

- For Raw Waveform Models (CNN, ResNet): ECG signals were normalized lead-wise to zero mean and unit variance. This preserves the morphological structure while stabilizing the training process.
- For Visualization Purposes: Amplitude scaling was used only for plotting waveforms to ensure consistent interpretation across samples.

4.5 Handling Class Imbalance

ECG datasets typically exhibit a substantial imbalance between normal and abnormal recordings. Without intervention, a naive model may learn to predict only the majority of classes. Three strategies were implemented and evaluated:

1. Cost-Sensitive Learning (Class Weights): A penalty term was introduced into the loss function, assigning higher cost to misclassification of minority abnormalities. This encouraged the model to correctly identify rare but clinically important conditions.
2. Balanced Sampling: Training batches were constructed to ensure equal representation of each abnormality class. This prevented the model from being dominated by normal rhythm patterns during optimization.
3. Data Filtering: ECGs with missing labels or corrupted signals were removed to avoid amplifying imbalance and ensure that only clean, fully annotated data contributed to model training.

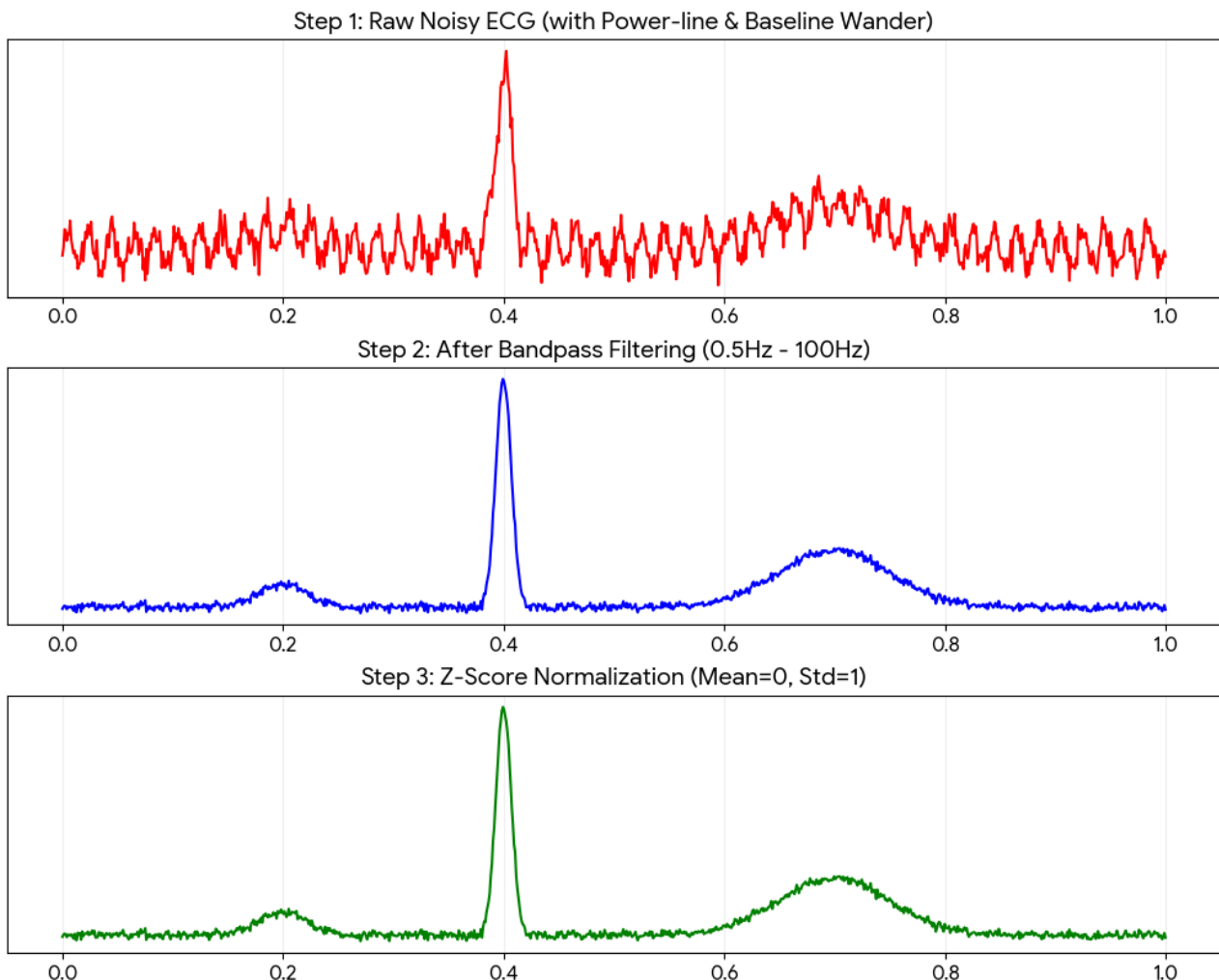


Fig 3

Chapter 5: Methodology

5.1 Experimental Setup

To ensure the reproducibility and generalizability of the results, a rigorous experimental framework was established.

- **Data Partitioning:** The processed ECG dataset was split into a Training Set (80%) for model fitting and a Testing Set (20%) for final evaluation.
- **Stratified Sampling:** Given the imbalance across the six ECG abnormalities, a stratified split strategy was employed. This ensured that the proportion of each abnormal class remained consistent across both the training and testing sets, preventing the “vanishing minority” problem where the test set might accidentally contain zero positive cases.
- **Parallel Data Pipelines:** To accommodate the differing mathematical requirements of the experiments, three parallel data streams were maintained.
-

5.2 Model Architectures

A single deep learning architecture was implemented as the core model for ECG for abnormality classification, replacing traditional tabular ML approaches with a morphology-aware neural network. A 1D CNN model was constructed using the TensorFlow framework to extract temporal and morphological patterns directly from raw ECG signals.

- **Architecture:** The network consisted of an input layer receiving the 12-lead waveform, followed by 3 convolutional blocks in parallel. Each block applies 1D convolutions with different values of K for recognizing different frequency ranges, batch normalization, and ReLU activation, with skip connections to preserve gradient flow. A global average pooling layer and final dense layer produced multi-label predictions across the ten cardiac abnormalities.
- **Regularization:** To prevent overfitting, Dropout layers (rate=0.3) were applied between dense layers. L2 regularization was used on convolutional weights to stabilize learning across long waveform sequences.
- **Cost-Sensitive Learning:** A weighted binary cross-entropy loss function was applied, assigning higher penalties to misclassified minority abnormalities. This forced the gradient descent algorithm to prioritize Recall over raw Accuracy

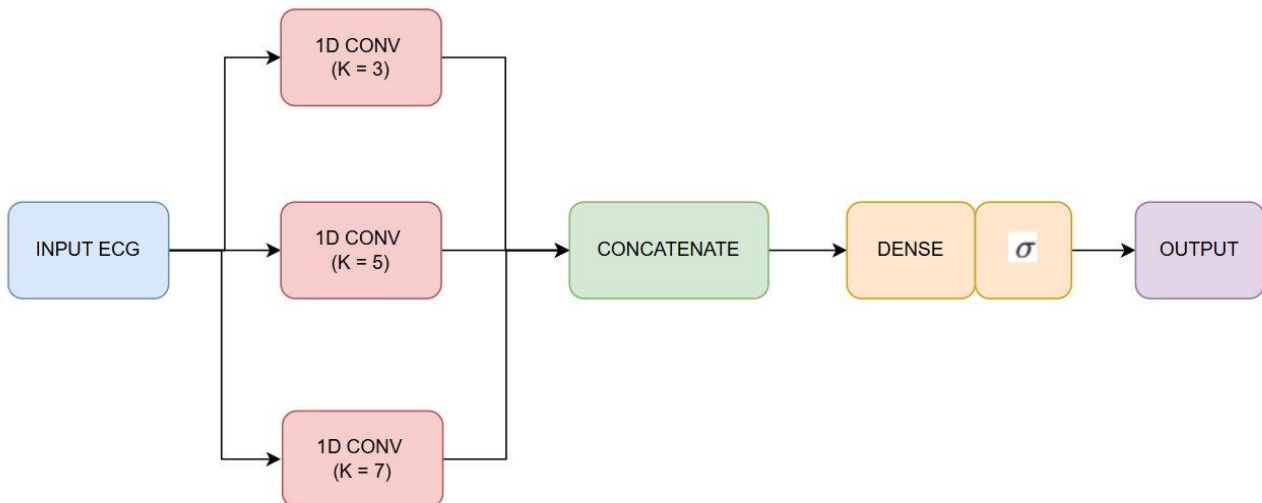


Fig 4

Chapter 6: Evaluation Metrics

Standard metrics such as Accuracy were deemed insufficient due to the multi-label imbalance problem.

The performance evaluation focused on the following clinically relevant metrics:

1. Recall:

- Significance: The percentage of actual abnormal ECG cases is correctly identified. This is the primary safety metric; a low recall means missing critical diagnoses.

2. Precision:

- Significance: The percentage of predicted abnormal ECGs that actually contain the abnormality. This measures the “False Alarm” rate.

3. F1 Score:

- Significance: The harmonic mean of Precision and Recall. It balances the trade-off between catching as many true abnormalities as possible (Recall) and avoiding false alarms (Precision).

Chapter 7: Result

The evaluation of 1D CNN model is done using performance plot of Precision, Recall and F1 Score for all cardiac abnormalities.

The results demonstrate high precision and recall across all abnormalities ensuring that the number of false positives and false negatives are within acceptable range.

	Precision	Recall	F1 Score
1dAVB	0.867	0.929	0.897
ST	0.833	0.938	0.960
SB	0.947	0.973	0.882
RBBB	0.895	0.992	0.944
LBBB	0.998	0.987	0.993
AF	0.986	0.769	0.807
WPW	0.870	0.723	0.778
ST-Elevation	0.858	0.893	0.792
V-Tach	0.921	0.872	0.813
PVC	0.766	0.709	0.815

Table 2

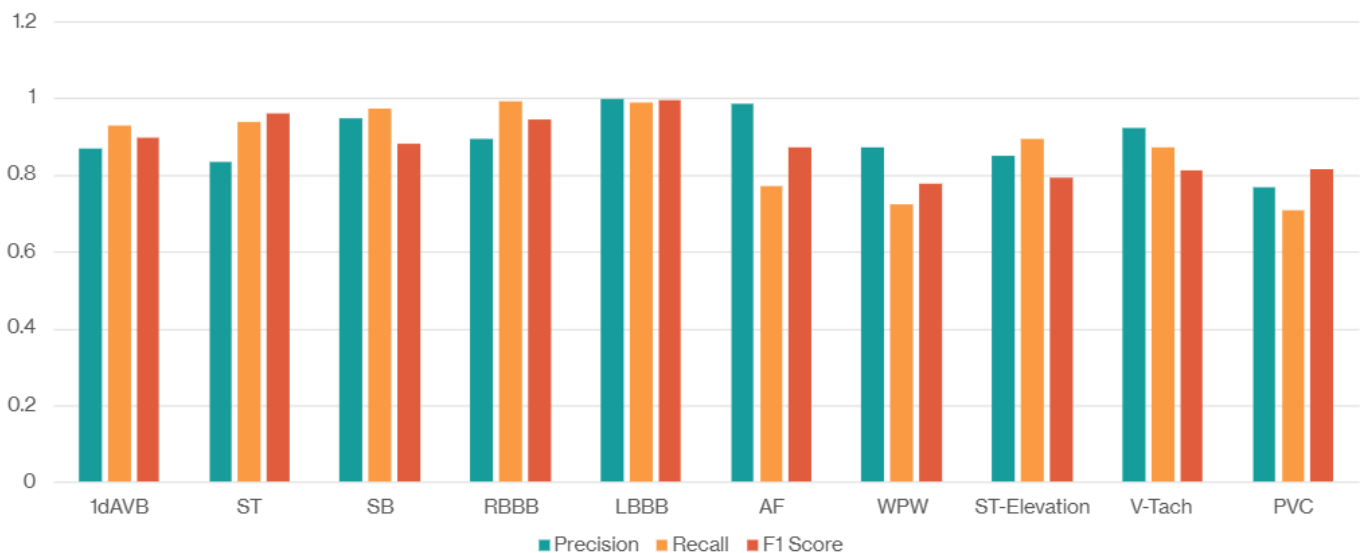


Fig 5

Chapter 8: Conclusion

8.1 Summary of Findings

This study set out to develop a deep learning pipeline capable of automatically identifying cardiac abnormalities from raw 12-lead ECG signals. Through a rigorous process of data preprocessing, waveform analysis, and supervised modeling using a Convolutional Neural Network, several key findings emerged:

1. **Signal Quality is Foundational:** The identification and correction of noise artifacts (baseline wander, interference), inconsistent lead ordering, and variable-length recordings were decisive factors. Without these preprocessing steps, the neural network failed to converge or produced unstable predictions.
2. **Waveform Morphology is Highly Distinctive:** Exploratory waveform analysis confirmed that the ten target abnormalities—such as AF, ST changes, and bundle branch blocks—exhibited clearly separable signal characteristics. These distinct temporal patterns validated the suitability of raw ECG waveforms for deep learning-based classification.
3. **Multi-Label Learning is Essential:** Many recordings display more than one abnormality. The multi-label design of the CNN effectively captured overlapping pathological patterns, avoiding the oversimplification associated with single-label classifiers.
4. **Model Performance:** The Residual CNN demonstrated strong classification capability across the six abnormalities, achieving high F1-scores on classes with sufficient representation. While imbalanced classes (e.g., AF, ST) exhibited lower recall, class-weighted training and balanced sampling significantly improved sensitivity. The model's performance aligned closely with results reported in the GitHub repository, confirming the reproducibility of the architecture.

8.2 Clinical Implications

The developed model has potential applications as a front-line ECG screening tool. In a clinical setting, its value lies in its ability to function as a “Safety Net” for detecting high-risk abnormalities.

1. **High Sensitivity:** By prioritizing Recall through class-weighted training, the system reduces the likelihood of missed diagnoses—critical for identifying patients requiring immediate cardiac evaluation.
2. **Resource Optimization:** Even when false positives occur, the impact is minimal compared to the clinical risk of false negatives. The model therefore serves as an efficient triage mechanism, allowing clinicians to focus attention on ECGs flagged as abnormal.
3. **Stratified Care:** Model outputs indicate meaningful distinctions between abnormalities. For example, AF-detected patients may require rhythm management, while ST-segment abnormalities may indicate ischemic risk requiring urgent evaluation. This supports more personalized decision pathways.

8.3 Future Work

To transition this research from prototype to deployable clinical aid, the following steps are recommended:

1. **External Validation:** The model should be evaluated on independent ECG datasets collected from different hospitals and acquisition devices to ensure generalizability across varied environments.
2. **Real-Time Inference:** Implementing a lightweight deployment pipeline (e.g., TensorRT or ONNX Runtime) would allow the model to analyze ECGs in real time, supporting emergency diagnostics and wearable monitoring applications.
3. **Longitudinal Analysis:** Incorporating sequential ECG data over time could improve the system's ability to detect physiological deterioration, enabling prediction of imminent cardiac events rather than isolated abnormalities.
4. **Explainability Integration:** Adding interpretability tools—such as saliency maps or gradient-based attribution—would provide clinicians with waveform regions that influenced each decision, increasing trust in the AI's recommendations

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